Continuous Probability distribution:

(range) region space

For every x belonging to the region continuous random variable X, we assign a real number $f(x)$ which satisfies the conditions:

(i) $f(x) \geq 0$

(ii) $\int_{-\infty}^{\infty} f(x)\,dx = 1$

Then, $f(x)$ is called a continuous probability function or probability density function (PDF).

If (a, b) is a subinterval of range space of X, then the prob. that x lies in (a,b) is defined to be the integral of $f(x)$ b/w a & b, i.e,

$$P(a \leq x \leq b) = \int_a^b f(x)\,dx.$$

Cummulative distribution function:

If X is a continuous random variable with P.D.F, $f(x)$, then the function $F(x)$, can be defined as $F(x) = P(X \leq x) = \int_{-\infty}^{x} f(x)\,dx$,

here $F(x)$ is cummulative distribution func^n (CDF) of $f(x)$, i.e; to write $F(x) = P(-\infty \leq X \leq x)$.

Remark:

(i) If x_r is any real no., the prob. of $x \geq x_r$, then $P(x \geq x_r) = \int_{x_r}^{\infty} f(x)\,dx$ also,

$$P(x < x_r) = \int_{-\infty}^{x_r} f(x)\,dx = 1 - P(x \geq x_r)$$

(i) Integral $-\infty$ to ∞ $f(x)\,dx = 1$, geometrically means that the area bounded by the curve $f(x)$ and x axis = unity (1).

(ii) $P(a \le x \le b)$ is the area bounded by $f(x)$ and $x = a$ and $x = b$.

Mean and variance of continuous prob. distribution:-

mean, $\mu = \int_{-\infty}^{\infty} x f(x)\,dx$

Variance, $\sigma^2 = \int_{-\infty}^{\infty} (x-\mu)^2 f(x)\,dx$

Examples:

1.) Check whether the following functions are ~~conti~~ pdf or not.

(i) $f(x) = \begin{cases} 2x, & -1 < x < 1 \\ 0 & \text{otherwise} \end{cases}$

Ans: when $x = -0.5$, $f(x) = -1$

$\Rightarrow f(x)$ is -ve, this means first cond^n not satisfied

(iv) $f(x) = \begin{cases} 2x & 0 < x \le 1 \\ 4-4x & 1 < x < 2 \\ 0 & \text{otherwise} \end{cases}$

Ans: For $x = 1.5$, $f(x) = -2$.

$\Rightarrow f(x)$ is -ve, first cond^n not satisfied

(iii) $f(x) = \begin{cases} |x|, & |x| \le 1 \\ 0 & \text{otherwise} \end{cases}$ $\quad |x| = \begin{cases} x & 0 < x < 1 \\ -x & -1 < x < 0 \end{cases}$

$\int_{-\infty}^{\infty} |x|\,dx$

$\int_{-1}^{0} -x\,dx + \int_{0}^{1} x\,dx$

$= 2\int_{0}^{1} x\,dx = 2\left[\frac{x^2}{2}\right]_0^1 = \left(\frac{1}{2}\right)(2) = 1$ //.

$\therefore f(x)$ is a pdf.

2.) Find the constant k such that $f(x) = \begin{cases} k x^2 & 0 < x < 3 \\ 0 & \text{otherwise} \end{cases}$ is a pdf. (i) Also find (ii) $P(1 < x < 2)$ (iii) $P(x \le 1)$ (iv) $P(x > 2)$ Also find mean & variance

Ans: By first condition, $k \ge 0$.

By second cond^n, $\int_{-\infty}^{\infty} f(x)\,dx = 1$.

$\Rightarrow \int_0^3 k x^2\,dx = 1$

$\Rightarrow k\int_0^3 x^2\,dx = 1 \Rightarrow k\left[\frac{x^3}{3}\right]_0^3 = 1$

$\Rightarrow k \times 9 = 1 \Rightarrow \boxed{k = 1/9}$

$\therefore$ pdf, $f(x) = \begin{cases} x^2/9 & 0 < x < 3 \\ 0 & \text{otherwise} \end{cases}$

$P(1 < x < 2) = \int_1^2 (1/9)x^2\,dx = \frac{1}{9}\left(\frac{x^3}{3}\right)_1^2$

$= \frac{1}{9}\left(\frac{8}{3} - \frac{1}{3}\right) = \frac{7}{27}$

$P(x \le 1) = \int_0^1 \frac{1}{9}x^2\,dx = \frac{1}{9}\left(\frac{x^3}{3}\right)_0^1 = \frac{1}{27}$ //

Exponential distribution

3) Find the cumulative dist. function for following pdf of random variable x.

(i) $f(x) = \begin{cases} 6x - 6x^2, & 0 \le x \le 1 \\ 0 & \text{otherwise} \end{cases}$

(ii) $f(x) = \begin{cases} x/4\, e^{-x/2} & 0 < x < \infty \\ 0 & \text{otherwise} \end{cases}$

Ans: (i) C.d.f, $F(x) = \int_{-\infty}^{x} f(x)\,dx$

$$F(x) = \int_{-\infty}^{0} f(x)\,dx + \int_{0}^{x} f(x)\,dx$$

$$= \int_{0}^{x} [6x - 6x^2]\,dx$$

$$= 6\frac{x^2}{2} - 6\frac{x^3}{3}$$

$$= 3x^2 - 2x^3 \qquad 0 \le x \le 1$$

(ii) C.d.f $F(x) = \int_{-\infty}^{x} f(x)\,dx$

$$= \int_{-\infty}^{0} f(x)\,dx + \int_{0}^{x} f(x)\,dx$$

$$= \int_{0}^{x} \left[\frac{x}{4} e^{-x/2}\right] dx \qquad (u \quad v)$$

$$= \frac{x\, e^{-x/2}}{1/2} + \frac{e^{-x/2}}{1/4}$$

$$= 2e^{-x/2}[2 - x] \qquad 0 < x < \infty$$

Exponential distribution

The continuous probability distribution having pdf, $f(x)$ given by

$$f(x) = \begin{cases} \alpha e^{-\alpha x}, & x \ge 0 \\ 0 & \text{otherwise}, \ \alpha > 0 \end{cases}$$

is known as the exponential distribution.

It is evident that, for every $\alpha > 0$, $f(x) \ge 0$ —(1)

To prove: $\int_{-\infty}^{\infty} f(x)\,dx = 1$

$$LHS = \int_{-\infty}^{\infty} f(x)\,dx$$

$$= 0 + \int_{0}^{\infty} f(x)\,dx$$

$$= \int_{0}^{\infty} \alpha e^{-\alpha x}\,dx$$

$$= \alpha \int_{0}^{\infty} e^{-\alpha x}\,dx = x\, e^{-\alpha x}$$

$$= \left[e^{-\alpha x}\right]_{0}^{\infty} = -[0 - 1]$$

Mean $\mu = 1/\alpha$; Variance, $\sigma^2 = 1/\alpha^2 \Rightarrow$ S.D. $= 1/\alpha$

1.) If x is an exponential variate, with mean 3, then find (i) $P(x>1)$ (ii) $P(x<3)$

Ans: (i) Mean, $\mu = 3$

$\Rightarrow 1/\alpha = 3$

$\Rightarrow \boxed{\alpha = 1/3}$

$\Rightarrow$ The pdf is $f(x) = \begin{cases} \frac{1}{3} e^{-x/3}, & x \geq 0 \\ 0 & \text{otherwise} \end{cases}$

(i) $P(x>1) = ?$

$$P(x>1) = \int_1^\infty \frac{1}{3} e^{-x/3}\, dx$$

$$= \frac{1}{3} \int_1^\infty e^{-x/3}\, dx$$

$$= \frac{1}{3} \times \frac{e^{-x/3}}{(-1/3)} \Big]_1^\infty$$

$$= -\left[e^{-x/3}\right]_1^\infty$$

$$= -[0 - e^{-1/3}]$$

$$= e^{-1/3} = 0.7165$$

(ii) $P(x<3) = \frac{1}{3} \int_0^3 e^{-x/3}\, dx$

$$= \frac{1}{3} \times \frac{\left[e^{-x/3}\right]_0^3}{[-1/3]}$$

$$= -\left[e^{-x/3}\right]_0^3 = -[e^{-1} - e^0]$$

$$= 1 - \frac{1}{e}$$

$$= 0.6321$$

2.) The kilometre run (in 1000's of kilometers) without any sort of problem in respect of certain vehicle is a random variable having pdf

$$f(x) = \begin{cases} \frac{1}{40} e^{-x/40} & x \geq 0 \\ 0 & \text{otherwise.} \end{cases}$$

Find the probability that the ~~pdf~~ vehicle is trouble free,

(i) Atleast for 25000 kilometers

(ii) Atmost for 25000 kilometers

(iii) between 16000 to 32000 kilometers.

Ans: Let x be the random variable representing kilometers in multiples of 1000, Regarding trouble free run by the vehicle

(i) $P(x \geq 25) = \int_{25}^\infty \frac{1}{40} e^{-x/40}\, dx$

$$= \frac{1}{40} \int_{25}^\infty e^{-x/40}\, dx$$

$$= \frac{1}{40} \times \left[\frac{e^{-x/40}}{(-1/40)}\right]_{25}^\infty$$

$$= -\left(e^{-x/40}\right)_{25}^\infty = -[0 - e^{-25/40}]$$

$P(x \geq 25) = e^{-25/40}$

$= 0.5352$

(ii) $P(x \leq 25) = \int_0^{25} \frac{1}{40} e^{-x/40}\, dx$

$= \frac{1}{40} \frac{\left[e^{-x/40}\right]_0^{25}}{[-1/40]}$

$= -\left[e^{-x/40}\right]_0^{25}$

$= -\left[e^{-25/40} - 1\right]$

$= 1 - e^{-25/40}$

$= 0.4647$

(iii) $P(16 \leq x \leq 32) = \int_{16}^{32} \frac{1}{40} e^{-x/40}\, dx$

$= \frac{1}{40} \frac{\left(e^{-x/40}\right)}{(-1/40)} \Big]_{16}^{32}$

$= -\left[e^{-x/40}\right]_{16}^{32}$

$= e^{-16/40} - e^{-32/40}$

$= 0.2209$

3) If x is an exponential variate with mean 5, evaluate (i) $P(0 < x < 1)$, (ii) $P(-\infty < x < 10)$ (iii) $P(x \leq 0 \text{ or } x \geq 1)$

Ans: As x is an exponential variate,

$$f(x) = \begin{cases} \alpha e^{-\alpha x} & x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

given: mean, $\mu = 5$

W.K.T $\mu = 1/\alpha$

$\Rightarrow 1/\alpha = 5 \Rightarrow \alpha = 1/5$

$$\therefore f(x) = \begin{cases} \frac{1}{5} e^{-x/5} & x \geq 0 \\ 0 & x < 0 \end{cases}$$

(i) $P(0 < x < 1) = \int_0^1 f(x)\, dx$

$= \int_0^1 \frac{1}{5} e^{-x/5}\, dx$

$= \frac{1}{5} \times \frac{e^{-x/5}}{(-1/5)} \Big]_0^1$

$= -e^{-x/5}\Big]_0^1 = -\left[e^{-1/5} - e^0\right]$

$= 1 - e^{-1/5} = 0.1812$

(ii) $P(-\infty < x < 10) = \int_{-\infty}^{10} f(x)\, dx$

$= \int_{-\infty}^{0} f(x)\, dx + \int_0^{10} f(x)\, dx$

$= 0 + \int_0^{10} \frac{1}{5} e^{-x/5}\, dx$

$= \frac{1}{5} \times \frac{e^{-x/5}}{(-1/5)} \Big]_0^{10}$

$= -\left[e^{-10/5} - e^0\right]$

$= 1 - e^{-2} = 0.864$

(iii) $P(x \le 0 \text{ or } x \ge 1)$

$= P(x \le 0) + P(x \ge 1)$

$= \int_{-\infty}^{0} f(x)\,dx + \int_{1}^{\infty} f(x)\,dx$

$= 0 + \int_{1}^{\infty} \frac{1}{5} e^{-x/5}\,dx$

$= \frac{1}{5} \times \frac{e^{-x/5}}{(-1/5)} \Big]_{1}^{\infty}$

$= -[0 - e^{-1/5}]$

$= e^{-1/5} = 0.818$

2) the length of telephone conversation in a booth has been an exponential distribution & found on average to be 5 minutes. find the probability that a random call made from this booth

(i) Ends less than 5 min

(ii) ends blw 5 to 10 min.

Ans: $\mu = \frac{1}{\alpha} = 5 \Rightarrow \alpha = 1/5$ x be length of conversation

$\therefore f(x) = \begin{cases} \frac{1}{5} e^{-x/5} & x \ge 0 \\ 0 & x < 0. \end{cases}$

(i) $P(x \le 5) = \int_{-\infty}^{5} f(x)\,dx$

$= \int_{-\infty}^{0} f(x)\,dx + \int_{0}^{5} f(x)\,dx$

$= 0 + \frac{1}{5} \int_{0}^{5} e^{-x/5}\,dx.$

$= \frac{1}{5} \times \frac{e^{-x/5}}{(-1/5)} \Big]_{0}^{5}$

$= -[e^{-x/5}]_{0}^{5}$

$= -[e^{-1} - 1] = 1 - e^{-1} = 0.632$

(ii) $P(5 \le x \le 10) = \int_{5}^{10} f(x)\,dx$

$= \frac{1}{5} \int_{5}^{10} e^{-x/5}\,dx$

$= \frac{1}{5} \times \frac{e^{-x/5}}{(-1/5)} \Big]_{5}^{10}$

$= -[e^{-10/5} - e^{-5/5}]$

$= -[e^{-2} - e^{-1}] = 0.232$

5) In a certain town the sales a day on a shop is exponentially distributed with an average sale amount to rupees 100 and net profit is 8%. find the probability that the net profit exceeds 30 Rs in two consecutive days

Ans: let x be the random variable which defines sales in a shop. As x is an exponential variate, we have

$f(x) = \begin{cases} \alpha e^{-\alpha x} & x \ge 0 \\ 0 & x < 0 \end{cases}$

the average sale is 100.

$\Rightarrow \mu = 100 = 1/\alpha \quad \alpha = \frac{1}{100}$

$\Rightarrow f(x) = \begin{cases} 100\, e^{-x/100} & x \ge 0. \\ 0 & x < 0 \end{cases}$

Let A be the amount for which profit is 8%.

$A \times 8\% = 30$

$\Rightarrow A = \frac{30 \times 100}{8} = 375$

Probab. of net profit exceeding 30 is

$$1 - P(X \leq 375)$$

$$= 1 - \int_0^{375} \frac{1}{100} e^{-x/100}\, dx$$

$$= 1 - \left[-(e^{-x/100})_0^{375} \right]$$

$$= 1 + e^{-x/100} \Big]_0^{375}$$

$$= 1 + \left[e^{-375/100} - 1 \right]$$

$$= e^{-375/100}$$

$$= 0.023$$

Prob. that profit exceeding 30 on ~~2 consecutive~~ next days

$$0.023 \times 0.023 = 5.29 \times 10^{-4}$$

Assignment – Collect 5 problems from old QP & write

Normal distribution

The continuous probability distribution having pdf, $f(x)$ given by

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

where $-\infty < x < \infty$

$-\infty < \mu < \infty$, $\sigma > 0$

is known as normal distribution.

mean $= \mu$, variance, $v = \sigma^2$

According to range of x, μ and σ, $f(x) \geq 0$

$$\int_{-\infty}^{\infty} f(x)\,dx = \int_{-\infty}^{\infty} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}\, dx$$

$$= \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{(x-\mu)^2}{2\sigma^2}}\, dx \quad \text{— ①}$$

$$= \frac{2\sigma}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-(x-\mu)^2}\, dx$$

Let $t^2 = \frac{(x-\mu)^2}{2\sigma^2}$

$$\Rightarrow t = \frac{(x-\mu)}{\sqrt{2}\sigma}$$

$$e^{-\frac{(x-\mu)^2}{2\sigma^2}} = e^{-(x-\mu)^2} \cdot e^{\frac{2\sigma^2}{2\sigma^2}}$$

$$\Rightarrow dt = \frac{dx}{\sqrt{2}\sigma}$$

$$\Rightarrow dx = \sqrt{2}\sigma\, dt \quad \text{— ②}$$

as x is from $-\infty$ to ∞, $t = -\infty$ to ∞

from ① & ②,

$$\int_{-\infty}^{\infty} f(x)\,dx = \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-t^2} dt \times \sqrt{2}\sigma$$

$$= \frac{\sqrt{2}}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-t^2} dt$$

$$= \frac{1}{\sqrt{\pi}} \int_{-\infty}^{\infty} e^{-t^2} dt = \frac{2}{\sqrt{\pi}} \int_0^{\infty} e^{-t^2} dt$$

$$\frac{2}{\sqrt{\pi}} \int_0^{\infty} e^{-t^2} dt = \frac{\sqrt{\pi}}{2}$$ by gamma func^n.

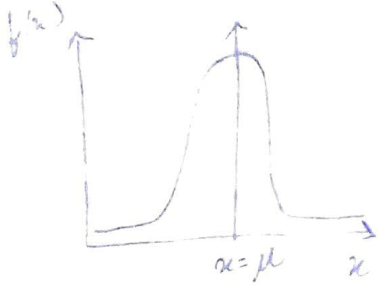
$$\Rightarrow \int f(x) = \frac{2}{\sqrt{\pi}} \times \frac{\sqrt{\pi}}{2} = 1$$

If $f(x)$ is a continuous prob. distribution

Standard normal distribution

$$P(a \le x \le b) = \frac{1}{\sigma\sqrt{2\pi}} \int_a^b e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx$$

Remark: the graph of prob. func^n $f(x)$ is a bell shaped curve, symmetrical about the line $x = \mu$, and is called the normal prob. curve.



The line $x = \mu$ divides the total area under the curve, i.e. $= 1$ into two equal parts. the area to the right as well as the left of $x = \mu$ is $\frac{1}{2}$.

We know that the probability $P(a \le x \le b)$ $= \int_a^b f(x)\,dx$. in normal dist.,

$$P(a \le x \le b) = \frac{1}{\sigma\sqrt{2\pi}} \int_a^b e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx$$

To simplify the above integrals, (1)

Let $z = \frac{x-\mu}{\sigma}$ (standard normal variate)

$x = \sigma z + \mu \Rightarrow dx = \sigma\,dz$

Let $z_1 = \frac{a-\mu}{\sigma}$, $z_2 = \frac{b-\mu}{\sigma}$ be the values of z corresponding to $x = a$ and $x = b$, respectively.

From (1),

$$P(z_1 \le z \le z_2) = \frac{1}{\sigma\sqrt{2\pi}} \int_{z_1}^{z_2} e^{-z^2/2} \sigma\,dz$$

$$= \frac{1}{\sqrt{2\pi}} \int_{z_1}^{z_2} e^{-\frac{z^2}{2}}\,dz$$

Let $f(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$ is called std^n normal

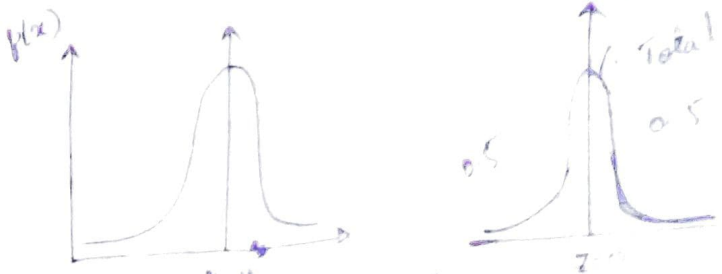
$z = \frac{x-\mu}{\sigma}$ (std. normal variate)

$dx = \sigma\,dz$

$z_1 = \frac{a-\mu}{\sigma}$, $z_2 = \frac{b-\mu}{\sigma}$, etc.

$$f(z) = \frac{1}{\sqrt{2\pi}} \int_{z_1}^{z_2} e^{-z^2/2}\,dz$$

If $\mu = 0$, $\sigma = 1$



Standard normal probab table

↳ use it to evaluate $P(z_1 \le z \le z_2)$ or $\phi(z)$

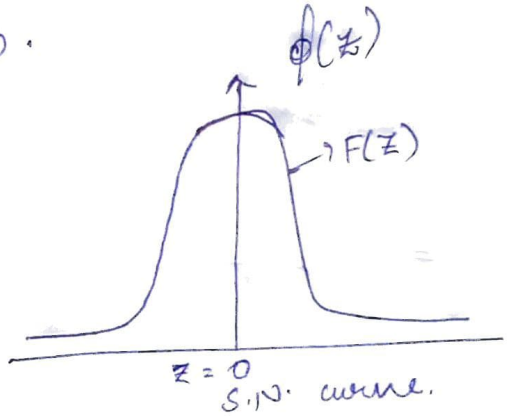
$$\phi(z) = \frac{1}{\sqrt{2\pi}} \int_0^z e^{-z^2/2}\,dz$$

Ex $\phi(1.23) = \frac{1}{\sqrt{2\pi}} \int_0^{1.23} e^{-z^2/2}\,dz$

$= 0.3907$

The particular case when $\mu = 0$ & $\sigma = 1$, then it is called ~~particular cond^{n}~~ of as standard normal distri

$F(z)$ is a standard normal curve, symmetrical about $z = 0$.



The integral

$$P(z_1 \le z \le z_2) = \frac{1}{\sqrt{2\pi}} \int_{z_1}^{z_2} e^{-z^2/2}\, dz$$

geometrically means that area bounded by S.N. curve $F(z)$ b/w $z = z_1$ & $z = z_2$.

Further in particular, if $z_1 = 0$ & $z_2 = z$, we have $\phi(z) = \frac{1}{\sqrt{2\pi}} \int_{0}^{z} e^{-z^2/2}\, dz$

This represents the area under S.N. curve from $z = 0$ & $z = z$

Some imp. results:

(i) $\int_{-\infty}^{\infty} \phi(z)\, dz = 1$ (ii) $\int_{-\infty}^{0} \phi(z)\, dz = \int_{0}^{\infty} \phi(z)\, dz = 0.5$

(iii) $P(-\infty < z < \infty) = 1$ (iv) $P(-\infty \le z \le 0) = P(0 \le z < \infty) = 0.5$

(v) $P(-\infty < z < z_1) = 0.5 + \int_{0}^{z_1} \phi(z)\, dz$,

i.e., $P(-\infty < z < 0) + P(0 < z < z_1)$

$P(z < z_1) = 0.5 + \phi(z_1)$

$P(z > z_1) = 0.5 - \phi(z_1)$